

Jupiter Database

Technical Reference & Exploratory Analysis

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Note: This document is a living reference. Update it as you learn more about the database. It serves two purposes: a day-to-day cheat sheet while building the system, and raw material for the Data chapter of the thesis report.

1. Database Overview

JupiterXL is Denmark's national database for groundwater and drinking water chemistry, maintained by the Geological Survey of Denmark and Greenland (GEUS). It is publicly available as a nightly export and serves citizens, municipalities, researchers, and regulatory authorities.

The version used in this project is the PCJupiterXL Plus export, downloaded on 26 March 2026. It was restored from a PostgreSQL custom-format backup (`pg_dump -Fc`) into a local PostgreSQL 16 instance.

Property	Value
Database format	PostgreSQL 16 custom backup (.backup)
Compressed size	3.7 GB (zip)
Uncompressed backup	19.9 GB
Schema name	pcjupiterxlplusviews
Total tables	276
Boreholes	477,105
Groundwater chemistry samples	508,413
Groundwater chemistry analyses	14,671,938
Chemical compounds tracked	3,026
Compounds with legal limits defined	1,217 (consumer), 675 (waterworks)
Data freshness	Nightly export — as recent as March 2026

2. Schema Structure

The 276 tables fall into a small number of functional groups. Understanding this grouping is essential before writing any queries or designing agent tools.

2.1 Table families

Family	Prefix	# Tables	Description
Core borehole & location	(none)	~8	borehole, intake, screen, locality. The spine of the database.
Groundwater chemistry	grw*	8	grwchemsample, grwchemanalysis, grwsoilsample, grwairanalysis etc.
Drinking water plant chemistry	plt*	8	Same structure as grw* but measured at treatment plant outlets.
Point-of-interest chemistry	poi*	8	Chemistry at contaminated sites and monitoring points.
Drinking water infrastructure	drw*	6	drwplant, drwplantintake, drwfir — water supply companies and plants.
Compound & limit lookups	(none)	5	compoundlist, compoundgroup, compoundgrouplist, limitlist.
Geology & lithology	(none)	~8	lithsamp, lithstra, biostra, minerals etc.
Water levels & hydrology	(none)	6	watlevel, wrrcatchment, wrrexport, wrrsupply.
Code lookup tables	code_*	~170	One table per coded field. e.g. code_752 = measurement units.
Data dictionary	jup*	6	juptable, jupfield, jupheadr — Jupiter's internal field-level metadata.

Note: ~170 of the 276 tables are code_* lookup tables. The actual data model has only ~30 meaningful tables.

2.2 The code_* lookup system

Every coded (numeric) field in Jupiter references a code_* table via a codetype number. Key codetypes for this project:

Codetype	Table	Description	Used in
752	code_752	Measurement units (mg/l, µg/l, pH etc.) — 292 units total	grwchemanalysis.unit, pltchemanalysis.unit, limitlist.unit
46	code_46	Physical dimension units (mm, cm, inches)	borhdiam.unit, screen.unit, casing.unit
221	compoundlist	Chemical compound numbers	grwchemanalysis.compoundno
17	code_17	Borehole purpose	borehole.purpose

3. Key Tables — Columns & Descriptions

3.1 borehole (477,105 rows)

The central table. Every other table links back here via boreholeid.

Column	Type	Description
boreholeid	numeric	Primary key. Used as FK in all related tables.
boreholeno	text	Human-readable DGU borehole number (e.g. '186. 358')
latitude / longitude	numeric	WGS84 decimal degrees. Some older records are NULL.
xutm32euref89 / yutm32euref89	numeric	UTM zone 32 EUREF89 — the modern Danish standard coordinate system.
municipal	text	Municipality name in Danish (e.g. 'København', 'Frederiksvark').
status	text	Borehole status code. 'R' = Registered/Active.
drilldepth	numeric	Total drilled depth in metres.
purpose	numeric	Coded field → code_17. Borehole purpose.
drilstdate / drilendate	timestamp	Drilling start and end dates.

3.2 grwchemsample (508,413 rows)

One row per sampling event at a borehole. Contains both boreholeid AND intakeid directly — the intake join can be skipped for basic queries.

Column	Type	Description
sampleid	numeric	Primary key. Referenced by grwchemanalysis.sampleid.
boreholeid	numeric	FK → borehole.boreholeid
intakeid	numeric	FK → intake.intakeid (depth interval within the borehole)
sampledate	timestamp	Date and time the sample was collected.
purpose	numeric	Coded field — reason for sampling.
watertype	numeric	Coded field — type of water sampled.
laboratory	numeric	Coded field → laboratory lookup.
qualitycontrol	numeric	Quality assurance status of the sample.

3.3 grwchemanalysis (14,671,938 rows)

The core chemistry table. One row per individual compound measurement. Nitrate, PFAS, pH, and all other chemical parameters live here.

Column	Type	Description
analysisid	numeric	Primary key.
sampleid	numeric	FK → grwchemsample.sampleid
compoundno	numeric	FK → compoundlist.compoundno. Which chemical was measured.
amount	numeric	The standardised measurement value. Always use this over reportedamount.
unit	numeric	FK → code_752. The unit of measurement (e.g. 1 = mg/l).
detectionlimit	numeric	Values below this are below detection. amount=0 often means <DL.
quantificationlimit	numeric	Lower bound for reliable quantification. Added 2024.
reportedamount / reportedunit	numeric	Raw lab values. Use amount/unit instead.
analysisdate	timestamp	Date the analysis was performed.
accreditedindicator	text	Whether performed by accredited laboratory.

3.4 compoundlist (3,026 rows)

Column	Type	Description
compoundno	numeric	Primary key. Used as FK in all analysis tables.
long_text	text	Full compound name in Danish (e.g. 'Nitrat').
short_text	text	Abbreviated name. Sometimes NULL.
casno	numeric	International CAS Registry Number.
grwunit	numeric	Default unit for groundwater measurements → code_752.
drwunit	numeric	Default unit for drinking water measurements → code_752.
molarweight	numeric	Molar weight in g/mol.

3.5 limitlist (1,347 rows)

Official Danish drinking water quality limits. Distinguishes between three points in the supply chain: waterworks (at treatment plant), pipeline network (distribution), and consumer (at the tap). For citizen-facing queries, consumer_max is the relevant threshold.

Column	Type	Description
compoundno	numeric	FK → compoundlist.compoundno
waterworks_max	numeric	Legal maximum at treatment plant outlet.
pipelinenetwork_max	numeric	Legal maximum in the distribution network.
consumer_max	numeric	Legal maximum at the consumer tap. Most relevant for citizen queries.
recommended_max	numeric	Recommended (non-binding) maximum. Very rarely populated (only 3 entries).
unit	numeric	FK → code_752. Unit of the limit values.

Note: 1,217 of 1,347 compounds have a `consumer_max` defined. 675 have a `waterworks_max`. Only 3 have a `recommended_max` — do not rely on this field.

4. Core Query Patterns

4.1 Standard chemistry query (with limit comparison)

The full production query: borehole → sample → analysis → compound → unit, with legal limit joined for automatic threshold flagging.

```

SELECT
    b.boreholeno,
    b.municipal,
    b.latitude,
    b.longitude,
    s.sampledate,
    a.amount,
    u.longtext AS unit,
    c.long_text AS compound,
    a.detectionlimit,
    l.consumer_max AS legal_limit,
    CASE WHEN l.consumer_max IS NOT NULL AND a.amount >
l.consumer_max
        THEN 'EXCEEDS LIMIT' ELSE 'OK' END AS status
FROM pcjupiterxlplusviews.borehole b
JOIN pcjupiterxlplusviews.grwchemsample s ON s.boreholeid =
b.boreholeid
JOIN pcjupiterxlplusviews.grwchemanalysis a ON a.sampleid =
s.sampleid
JOIN pcjupiterxlplusviews.compoundlist c ON c.compoundno =
a.compoundno
JOIN pcjupiterxlplusviews.code_752 u ON u.code = a.unit
LEFT JOIN pcjupiterxlplusviews.limitlist l ON l.compoundno =
a.compoundno

```

```

WHERE a.compoundno = 246
AND b.municipal = 'Frederiksvaerk'
AND s.sampledate >= '2020-01-01'
ORDER BY s.sampledate DESC
LIMIT 50;

```

4.2 Compound lookup by name

```

SELECT compoundno, long_text, short_text, casno, grwunit
FROM pcjupiterxlplusviews.compoundlist
WHERE long_text ILIKE '%nitrat%'
      OR short_text ILIKE '%nitrat%';

```

4.3 PFAS group query

```

SELECT c.compoundno, c.long_text
FROM pcjupiterxlplusviews.compoundlist c
JOIN pcjupiterxlplusviews.compoundgroup cg ON cg.compoundno =
c.compoundno
WHERE cg.compoundgroupno = 110;

```

4.4 Geographic proximity (Haversine)

```

SELECT boreholeid, boreholeno, municipal, latitude, longitude,
111.045 * DEGREES(ACOS(LEAST(1.0,
COS(RADIANS(b.latitude)) * COS(RADIANS(55.68)) *
COS(RADIANS(b.longitude) - RADIANS(12.57)) +
SIN(RADIANS(b.latitude)) * SIN(RADIANS(55.68))
))) AS distance_km
FROM pcjupiterxlplusviews.borehole b
WHERE latitude IS NOT NULL
HAVING distance_km < 5
ORDER BY distance_km;

```

5. Legal Limits Quick Reference

Verified against limitlist table. For use in agent responses when flagging exceedances.

Compound	compoundno	Consumer limit	Unit	Notes
Nitrat	246	50	mg/l	EU Drinking Water Directive. Real exceedance found: borehole 186.358,

Compound	compoundno	Consumer limit	Unit	Notes
				Frederiksvaerk, 56 mg/l (March 2026).
Nitrat-N	247	11.289	mg/l	Same compound, expressed as nitrogen content rather than NO3.
PFAS (sum: PFOA+PFOS+PFNA+PFHxS)	group 110	0.002	µg/l	Strictest PFAS limit. Applies to 4 specific compounds combined.
PFAS (individual, most)	group 110	0.1	µg/l	Individual limit for most other PFAS compounds in group 110.
Sum af PFAS, 12 stoffer	group 110	0.1	µg/l	EU aggregate sum limit for 12 PFAS compounds.

6. Important Gotchas & Data Quality Notes

Issue	Detail	How to handle
amount=0 ≠ zero	Historical records use amount=0 for below-detection, not true zero.	Check detectionlimit. If amount ≤ detectionlimit, report as below detection.
NULL coordinates	Some boreholes lack lat/lon, especially older ones.	Filter WHERE latitude IS NOT NULL for geographic queries.
Danish municipality names	Special characters: 'København', 'Ålbæk', 'Frederiksvark'.	Use ILIKE for matching. Agent must handle name variants.
amount vs reported*	Both exist in grwchemanalysis. reported* = raw lab submission.	Always use amount and unit, never reportedamount/reportedunit.
Danish compound names	All names in compoundlist are Danish.	Agent prompt must map English queries to Danish names.
unit is numeric	code_752.code is numeric. No text casting needed.	JOIN on u.code = a.unit directly.
Schema prefix required	All tables are in pcjupiterxlplusviews, not public.	All queries must prefix: pcjupiterxlplusviews.tablename
14.6M analysis rows	Unfiltered queries on grwchemanalysis will be slow.	Always filter by compoundno and date range first.
PFAS two-tier limits	PFAS has two different limit levels depending on compound.	Always join limitlist per compound; do not assume a single PFAS limit.

7. Built-in Data Dictionary (jup* tables)

Jupiter contains its own internal metadata. The jupfield table is particularly useful — it maps every column to its codetype, enabling programmatic schema descriptions for the agent system prompt.

Table	Description	Key use
juptable	One row per data table. Contains English and original Danish Oracle name.	Maps table names to descriptions.
jupfield	One row per column per table. Contains codetype (which code_* table it references) and original Danish column name.	Programmatically determine join tables for any coded field.
jupheadr / jupheade	Header/heading definitions.	Less relevant for query generation.
juphiera	Table hierarchy definitions.	Understanding parent-child table relationships.

```
-- Example: find which code table decodes grwchemanalysis.unit
SELECT tablename, columnname, codetype
FROM pcjupiterxlplusviews.jupfield
WHERE tablename = 'GRWCHEMANALYSIS' AND columnname = 'UNIT';
-- Returns codetype = 752, confirming code_752 is the unit lookup
table.
```

8. Local Setup Reference

Parameter	Value
WSL2 host	GEUS3211W10WS
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
Database name	jupiterxl
DB user	pgmcp
DB password	pgmcp_dev
Schema	pcjupiterxlplusviews
Backup file	~/Thesis/data/pcjupiterxlpluspg_xl.backup
Restore command	pg_restore -h localhost -U pgmcp -d jupiterxl --no-owner --role=pgmcp -v pcjupiterxlpluspg_xl.backup
Start PostgreSQL	sudo service postgresql start
Connect to DB	psql -h localhost -U pgmcp -d jupiterxl

9. Next Steps

- Adapt database.py from the toy project to connect to jupiterxl with schema pcjupiterxlplusviews
- Update server.py MCP tools: add search_compound(name), get_compound_group(group_name), and check_limit(compoundno) tools
- Add limitlist to the agent context so it can automatically flag exceedances in responses
- Investigate drwplant and drwplantintake tables for drinking water plant coverage
- Check whether PostGIS is available for proper geospatial queries
- Profile grwchemanalysis query performance — 14.6M rows may need indexes on compoundno + sampleid
- Design the system prompt with: schema prefix, Danish compound names, limit awareness, and detection limit handling